5.16 Construction of a Square

Let us now construct a square when (i) its side is given and (ii) its diagonal is given

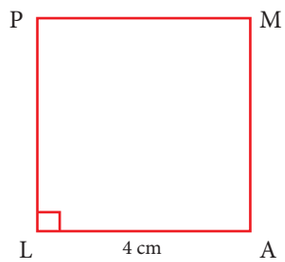
5.16.1 Construction of a square when its side is given

Example 5.40

Construct a square LAMP of side 4 cm. Also find its area.

Solution:

Given: side = 4 cm



Rough diagram

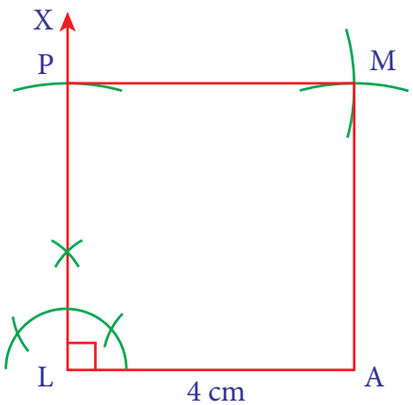


Fig. 5.59

Steps:

1. Draw a line segment LA = 4 cm.
2. At L, construct $LX \perp LA$.
3. With L as centre, draw an arc of radius 4 cm and let it cut LX at P.
4. With A and P as centres, draw arcs of radius 4 cm each and let them cut at M.
5. Join AM and PM. LAMP is the required square.

Calculation of Area:

Area of square LAMP $= a^2$ *sq.units*

$= 4 \times 4 = 16$ *sq.cm*

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